

Assignment – 9.5

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Task 1 – String Utilities Function

PROMPT :

```
#def reverse_string(text):
```

```
#return text[::-1]
```

#Write documentation in:(a) Docstring (b) Inline comments (c) Google-style documentation

CODE :

```
AI-9.5.py > ...
1  #def reverse_string(text):
2  #return text[::-1]
3  #Write documentation in:(a) Docstring (b) Inline comments (c) Google-style documentation
4  # (a) Docstring
5  def reverse_string(text):
6      """
7      Reverses the given string.
8
9      Args:
10         text (str): The string to be reversed.
11
12     Returns:
13         str: The reversed string.
14     """
15     return text[::-1]
16 # (b) Inline comments
17 def reverse_string(text):
18     # Reverse the string using slicing
19     return text[::-1]
20 # (c) Google-style documentation
21 def reverse_string(text):
22     """
23     Reverses the given string.
24
25     Args:
26         text (str): The string to be reversed.
27
28     Returns:
29         str: The reversed string.
30     """
31     return text[::-1]
32 #Compare the three documentation styles.
33 # The docstring provides a clear and concise description of the function, its parameters, and its return value.
34 # It is a standard way to document functions in Python and can be easily accessed using the help() function.
35 # The inline comments provide a brief explanation of the code within the function.
36 # They are useful for explaining specific lines of code or logic but may not provide a comprehensive overview of the function's purpose and usage.
37 # The Google-style documentation is similar to the docstring but follows a specific format that is widely used in the industry.
38 # It includes sections for Args and Returns, making it easy to understand the function's inputs and outputs at a glance.
39 # This style is particularly beneficial for larger projects or when collaborating with other developers, as it provides a consistent and structured way to document code.
40
```

JUSTIFICATION :

Google-style documentation is most suitable for a utility-based string library because it clearly defines parameters and return values in a structured format. It improves readability and supports automatic documentation generation tools. It is widely used in professional development environments. Therefore, it enhances maintainability and scalability of utility libraries.

Task 2 – Password Strength Checker

PROMPT :

```
#def check_strength(password):
```

#return len(password) >= 8

#Write documentation in:(a) Docstring (b) Inline comments (c) Google-style documentation

CODE :

```
42 #def check_strength(password):
43 #return len(password) >= 8
44 #Write documentation in:(a) Docstring (b) Inline comments (c) Google-style documentation
45 # (a) Docstring
46 def check_strength(password):
47     """
48     Checks the strength of a password based on its length.
49
50     Args:
51         password (str): The password to be checked.
52
53     Returns:
54         bool: True if the password is strong (length >= 8), False otherwise.
55     """
56     return len(password) >= 8
57 # (b) Inline comments
58 def check_strength(password):
59     # Check if the length of the password is greater than or equal to 8
60     return len(password) >= 8
61 # (c) Google-style documentation
62 def check_strength(password):
63     """
64     Checks the strength of a password based on its length.
65
66     Args:
67         password (str): The password to be checked.
68
69     Returns:
70         bool: True if the password is strong (length >= 8), False otherwise.
71     """
72     return len(password) >= 8
73 #Compare the three documentation styles.
74 # The docstring provides a clear and concise description of the function, its parameters, and its return value.
75 # It is a standard way to document functions in Python and can be easily accessed using the help() function.
76 # The inline comments provide a brief explanation of the code within the function.
77 # They are useful for explaining specific lines of code or logic but may not provide a comprehensive overview of the function's purpose and usage.
78 # The Google-style documentation is similar to the docstring but follows a specific format that is widely
79 # used in the industry. It includes sections for Args and Returns, making it easy to understand the function's inputs and outputs at a glance.
80 # This style is particularly beneficial for larger projects or when collaborating with other developers, as it provides a consistent and structured way to document code.
```

JUSTIFICATION :

For security-related code, structured documentation like Google-style is preferred because it clearly explains validation logic and return behavior. Security functions require precise parameter and output descriptions to avoid misuse. Proper documentation also helps in auditing and maintenance. Hence, structured documentation improves reliability and clarity.

Task 3 – Math Utilities Module

PROMPT :

#Create a module math_utils.py with functions: square(n), cube(n), factorial(n)

#Write documentation in:(a) Docstring (b) Inline comments (c) Google-style documentation

CODE :

```

126 def square(n):
127     # Return the square of the input number
128     return n * n
129 def cube(n):
130     # Return the cube of the input number
131     return n * n * n
132 def factorial(n):
133     # Check if the input number is negative
134     if n < 0:
135         raise ValueError("Input must be a non-negative integer.")
136     # Return 1 for factorial of 0 and 1
137     elif n == 0 or n == 1:
138         return 1
139     else:
140         result = 1
141         # Calculate factorial using a loop
142         for i in range(2, n + 1):
143             result *= i
144         return result
145 # (c) Google-style documentation
146 def square(n):
147     """
148     Returns the square of a number.
149
150     Args:
151         n (int or float): The number to be squared.
152
153     Returns:
154         int or float: The square of the input number.
155     """
156     return n * n
157 def cube(n):
158     """
159     Returns the cube of a number.
160     Args:
161         n (int or float): The number to be cubed.
162     Returns:
163         int or float: The cube of the input number.
164     """
165     return n * n * n
166 def factorial(n):
167     """
168     Returns the factorial of a number.

```

```

83
84 #Create a module math_utils.py with functions: square(n), cube(n), factorial(n)
85 #Write documentation in:(a) Docstring (b) Inline comments (c) Google-style documentation
86 # math_utils.py
87 # (a) Docstring
88 def square(n):
89     """
90     Returns the square of a number.
91
92     Args:
93         n (int or float): The number to be squared.
94
95     Returns:
96         int or float: The square of the input number.
97     """
98     return n * n
99 def cube(n):
100     """
101     Returns the cube of a number.
102     Args:
103         n (int or float): The number to be cubed.
104     Returns:
105         int or float: The cube of the input number.
106     """
107     return n * n * n
108 def factorial(n):
109     """
110     Returns the factorial of a number.
111     Args:
112         n (int): The number to calculate the factorial of. Must be a non-negative integer.
113     Returns:
114         int: The factorial of the input number.
115     """
116     if n < 0:
117         raise ValueError("Input must be a non-negative integer.")
118     elif n == 0 or n == 1:
119         return 1
120     else:
121         result = 1
122         for i in range(2, n + 1):
123             result *= i
124         return result
125 # (b) Inline comments

```

```

156     return n * n
157 def cube(n):
158     """
159     Returns the cube of a number.
160     Args:
161         n (int or float): The number to be cubed.
162     Returns:
163         int or float: The cube of the input number.
164     """
165     return n * n * n
166 def factorial(n):
167     """
168     Returns the factorial of a number.
169     Args:
170         n (int): The number to calculate the factorial of. Must be a non-negative integer.
171     Returns:
172         int: The factorial of the input number.
173     """
174     if n < 0:
175         raise ValueError("Input must be a non-negative integer.")
176     elif n == 0 or n == 1:
177         return 1
178     else:
179         result = 1
180         for i in range(2, n + 1):
181             result *= i
182         return result
183
184 #Compare the three documentation styles.
185 # The docstring provides a clear and concise description of each function, its parameters, and its return value.
186 # It is a standard way to document functions in Python and can be easily accessed using the help() function.
187 # The inline comments provide a brief explanation of the code within each function.
188 # They are useful for explaining specific lines of code or logic but may not provide a comprehensive overview of the function's purpose and usage.
189 # The Google-style documentation is similar to the docstring but follows a specific format that is widely
190 # used in the industry. It includes sections for Args and Returns, making it easy to understand each function's inputs and outputs at a glance.
191 # This style is particularly beneficial for larger projects or when collaborating with other developers, as it provides a consistent and structured way to document code.

```

JUSTIFICATION :

Documenting mathematical functions improves clarity about input constraints and expected outputs. It helps prevent misuse, especially in functions like factorial where invalid inputs may cause errors. Automatically generated docstrings save time and maintain consistency. Exporting HTML documentation improves accessibility and professional presentation.

Task 4 – Attendance Management Module

PROMPT :

#Create a module attendance.py with functions: mark_present(student) , mark_absent(student) , get_attendance(student)

#Write documentation in:(a) Docstring (b) Inline comments (c) Google-style documentation

CODE :

```

237         return self.attendance_record.get(student, "No record found")
238     # (c) Google-style documentation
239     class Attendance:
240         """
241         A class to manage student attendance.
242
243         Methods:
244         mark_present(student): Marks a student as present.
245         mark_absent(student): Marks a student as absent.
246         get_attendance(student): Returns the attendance status of a student.
247         """
248         def __init__(self):
249             """Initializes an empty attendance record."""
250             self.attendance_record = {}
251
252         def mark_present(self, student):
253             """Marks a student as present.
254
255             Args:
256             | student (str): The name of the student to be marked as present.
257             """
258             self.attendance_record[student] = "Present"
259
260         def mark_absent(self, student):
261             """Marks a student as absent.
262
263             Args:
264             | student (str): The name of the student to be marked as absent.
265             """
266             self.attendance_record[student] = "Absent"
267
268         def get_attendance(self, student):
269             """Returns the attendance status of a student.
270
271             Args:
272             | student (str): The name of the student whose attendance status is to be retrieved.
273
274             Returns:
275             | str: The attendance status of the student, or a message if no record is found.
276             """
277             return self.attendance_record.get(student, "No record found")
278     # Compare the three documentation styles.
279     # The docstring provides a clear and concise description of the class and its methods, including their parameters and return values.

```

```

194 # Create a module attendance.py with functions: mark_present(student), mark_absent(student), get_attendance(student)
195 # Write documentation in: (a) Docstring (b) Inline comments (c) Google-style documentation
196 # attendance.py
197 # (a) Docstring
198 class Attendance:
199     """
200     A class to manage student attendance.
201
202     Methods:
203     mark_present(student): Marks a student as present.
204     mark_absent(student): Marks a student as absent.
205     get_attendance(student): Returns the attendance status of a student.
206     """
207     def __init__(self):
208         self.attendance_record = {}
209
210     def mark_present(self, student):
211         """Marks a student as present."""
212         self.attendance_record[student] = "Present"
213
214     def mark_absent(self, student):
215         """Marks a student as absent."""
216         self.attendance_record[student] = "Absent"
217
218     def get_attendance(self, student):
219         """Returns the attendance status of a student."""
220         return self.attendance_record.get(student, "No record found")
221 # (b) Inline comments
222 class Attendance:
223     def __init__(self):
224         # Initialize an empty dictionary to store attendance records
225         self.attendance_record = {}
226
227     def mark_present(self, student):
228         # Mark the student as present in the attendance record
229         self.attendance_record[student] = "Present"
230
231     def mark_absent(self, student):
232         # Mark the student as absent in the attendance record
233         self.attendance_record[student] = "Absent"
234
235     def get_attendance(self, student):
236         # Return the attendance status of the student, or a message if no record is found

```

```

259
260
261 def mark_absent(self, student):
262     """Marks a student as absent.
263
264     Args:
265         student (str): The name of the student to be marked as absent.
266     """
267     self.attendance_record[student] = "Absent"
268
269 def get_attendance(self, student):
270     """Returns the attendance status of a student.
271
272     Args:
273         student (str): The name of the student whose attendance status is to be retrieved.
274
275     Returns:
276         str: The attendance status of the student, or a message if no record is found.
277     """
278     return self.attendance_record.get(student, "No record found")
279
280 #Compare the three documentation styles.
281 # The docstring provides a clear and concise description of the class and its methods, including their parameters and return values.
282 # It is a standard way to document classes and methods in Python and can be easily accessed using the help() function.
283 # The inline comments provide a brief explanation of the code within each method.
284 # They are useful for explaining specific lines of code or logic but may not provide a comprehensive overview of the class's purpose and usage.
285 # The Google-style documentation is similar to the docstring but follows a specific format that is widely used in the industry. It includes sections for Args and Ret
286 # This style is particularly beneficial for larger projects or when collaborating with other developers, as it provides a consistent and structured way to document c

```

JUSTIFICATION :

Attendance systems require clear documentation to define function behavior and data handling. Proper docstrings improve maintainability and team collaboration. Generated documentation allows easy viewing in terminal and browser. This ensures better usability and transparency in academic or organizational systems.

Task 5 – File Handling Function

PROMPT :

```
#def read_file(filename):
```

```
#with open(filename, 'r') as f:
```

```
#return f.read()
```

#Write documentation in:(a) Docstring (b) Inline comments (c) Google-style documentation

CODE :

```

288 #def read_file(filename):
289 #with open(filename, 'r') as f:
290 #return f.read()
291 #Write documentation in:(a) Docstring (b) Inline comments (c) Google-style documentation
292 # (a) Docstring
293 def read_file(filename):
294     """
295     Reads the contents of a file and returns it as a string.
296
297     Args:
298         filename (str): The name of the file to be read.
299     Returns:
300         str: The contents of the file.
301     """
302     with open(filename, 'r') as f:
303         return f.read()
304 # (b) Inline comments
305 def read_file(filename):
306     # Open the file in read mode
307     with open(filename, 'r') as f:
308         # Read the contents of the file and return it
309         return f.read()
310 # (c) Google-style documentation
311 def read_file(filename):
312     """
313     Reads the contents of a file and returns it as a string.
314
315     Args:
316         filename (str): The name of the file to be read.
317     Returns:
318         str: The contents of the file.
319     """
320     with open(filename, 'r') as f:
321         return f.read()
322 #Compare the three documentation styles.
323 # The docstring provides a clear and concise description of the function, its parameters, and its return value.
324 # It is a standard way to document functions in Python and can be easily accessed using the help() function.
325 # The inline comments provide a brief explanation of the code within the function.
326 # They are useful for explaining specific lines of code or logic but may not provide a comprehensive overview of the function's purpose and usage.
327 # The Google-style documentation is similar to the docstring but follows a specific format that is widely used in the industry.
328 # It includes sections for Args and Returns, making it easy to understand the function's inputs and outputs at a glance.
329 # This style is particularly beneficial for larger projects or when collaborating with other developers, as it provides a consistent and structured way to document code.

```

JUSTIFICATION :

Google-style documentation best explains exception handling because it clearly defines raised exceptions and return values. File operations are prone to errors like `FileNotFoundError`, so structured documentation improves understanding. It helps developers anticipate edge cases. Therefore, it enhances robustness and maintainability of file-handling code.